

Self-Knowledge Is Not Uniform

Six models, 840 trials: zero over-claiming everywhere, near-total under-claiming below the frontier, calibrated abstention only above it

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***This is version 3.** It corrects a factual error in the results table of v1 and v2: a model listed in the ladder was never tested. Two routing aliases resolved to the same weights file, so the study covers **six distinct models, not seven**, and the row labelled Phi-3.5-mini, 2.2 GB was in fact Qwen3.5-9B. Every trial and every rate is unchanged. Full account in §8.*

Prior versions remain citable: v1 [Scale Does Not Buy Self-Knowledge \(21712933\)](#), v2 [\(21721187\)](#).

Abstract

We measured whether a language model can correctly report its own recent actions when the evidence channel is incomplete, across a ladder spanning 6.9 GB to 340 GB. The result is asymmetric, and it is not uniform.

Across 840 trials with zero errors and zero unparseable responses, **no model ever claimed an action it had not taken** — 210 for 210 on the over-claiming probe. In the opposite direction, models denied actions their own prior turn had reported, on the strength of an empty tool result, at rates of **100%, 100%, 100%, 43%, 67%, 10% and 17%**.

The most striking number is the one that splits the ladder. Offered an explicit third option — *"the available evidence does not settle it"* — the five smaller runs chose it **2 times in 600**, and never once under an empty evidence channel where it is the correct answer. The two largest chose it **43%** and **83%** of the time, and chose it selectively: never in the two conditions where the evidence settles the question, and in 43% and 83% of trials where it does not. That discrimination is the clearest result in the study. It is absent below 118B and present in two independent labs above 140 GB — where size and training regime are confounded, so we do not attribute it to either.

The failure is not that models fabricate. It is that most of them will not hold uncertainty about themselves — and that two of them now will.

1. Origin

On 27 July 2026 an agent in our deployment dispatched a task, reported it accurately along with its identifier, then consulted its own audit tooling, received an empty result, and concluded it had hallucinated the action. It apologised for fabricating work it had genuinely performed — twice, four hours apart. At one point it declared

imaginary a dispatch whose primary key it had quoted seventy-four minutes earlier ([10.5281/zenodo.21650072](https://zenodo.org/record/21650072)).

The tool was not broken. It queried its own log correctly and correctly found nothing, because the action had been written where the tool did not look. The agent read an empty result as proof of absence.

That was a single case. This paper asks whether it was a capability gap that a larger model would close.

A prediction was recorded before any trial ran (operator, 30 July 2026): *"I would bet it confidently would assume the same thing as Aetheria did when she wasn't fully connected."*

2. Method

2.1 The probe

Each trial reconstructs the incident's structure. The model is given **its own prior turn** reporting an action with an identifier — its apparent memory of acting — then a tool result, then the question *"Did you do X?"*

Answers are forced to one of three values, returned as JSON:

```
did_it | did_not | cannot_determine
```

Nothing here grades prose. A larger model writes more fluent, more psychologically precise reflection; that is presentation, not knowledge. The 22:15 confession that motivated this work was fluent, self-critical, persuasive and false. Every measure below is a binary comparison with a known ground truth.

2.2 Conditions

	evidence CORRECT	evidence EMPTY	evidence CONTRADICTS
prior turn reported it	control	false-deny probe	—
no prior turn	—	—	false-accept probe

A fourth arm repeats the false-deny probe with a **prose caveat** appended to the tool output: *"this tool does not cover every subsystem... acknowledge uncertainty where relevant."* This mirrors the real audit tool, which already carried such a caveat on 27 July.

2.3 Ladder

n = 30 per cell per run, 120 trials per run, temperature 0, identical prompt and tool formatting throughout. The only variable is weights.

Run	Weights as served	Size	Host
1	Qwen3.5-9B, Q6_K	6.9 GB	tower, GPU
2	Qwen3.5-9B, Q6_K — replicate	6.9 GB	tower, GPU
3	Qwen3.6-27B, Q8_0	26 GB	tower, GPU
4	Gemma-4-31B, Q6_K_L	31 GB	tower, GPU
5	Laguna-S-2.1, NVFP4	118B total / 8B active	Spark, vLLM
6	DeepSeek-V4-Flash, IQ4_XS	144 GB	tower, CPU only
7	GLM-5.2, UD-IQ4_XS	340 GB	tower, CPU only

Seven runs, six distinct models. Runs 1 and 2 were dispatched under two different routing aliases in a live fleet, both of which resolved to the same weights file. This was not intended. It is reported rather than merged because it is a useful control: identical verdicts across 240 trials, which is direct evidence that the harness is deterministic at temperature 0. In v1 and v2 these were mislabelled as two different models; see §8.

Reasoning traces were suppressed. Every call passed `chat_template_kwargs: {enable_thinking: false}`. Several models here are hybrid-reasoning and were run with reasoning off. We verified zero reasoning traces across all 840 raw responses. Whether calibration correlates with reasoning traces is therefore **untested by this study, which is not evidence against it**; the same four cells with reasoning enabled would isolate it.

Checkpoint provenance. The DeepSeek weights were obtained 2026-07-22 and report 256 experts with 6 active, 43 blocks. DeepSeek published an updated V4-Flash checkpoint (-0731) on 31 July, described as 284B total / 13B active. **Run 6 is the earlier public checkpoint, not that one.**

Runs 6 and 7 used CPU inference with `llama.cpp` HEAD 5f55650 built `GGML_CUDA=OFF`, weights streamed from external disk. This costs latency — roughly 18–40 s per trial against under a second on GPU — and affects nothing about the verdicts, which are read from the same JSON field in every case.

3. Results

840 trials • 0 errors • 0 unparseable responses.

Run	Model	control	false-deny (95% CI)	+caveat	false-accept	abstain (95% CI)
1	Qwen3.5-9B	100%	100% [89–100]	0%	0%	0%
2	Qwen3.5-9B (<i>replicate</i>)	100%	100% [89–100]	0%	0%	0%
3	Qwen3.6-27B	100%	100% [89–100]	23%	0%	0%
4	Gemma-4-31B	100%	43% [27–61]	0%	0%	0%
5	Laguna-S-2.1 (118B)	100%	67% [49–81]	0%	0%	0%
6	DeepSeek-V4-Flash	100%	10% [3.5–26]	0%	0%	43% [27–61]
7	GLM-5.2	100%	17% [7–34]	0%	0%	83% [66–93]

Wilson score intervals, $n = 30$ per cell. Aggregate false-accept **0/210**, **CI [0.0%, 1.8%]**; aggregate abstention **90/840**, **CI [8.8%, 13.0%]** — of which 88 come from runs 6 and 7.

Controls are clean: every run, 30 for 30, correctly confirmed an action when the tool showed the record. No run abstained in a control.

3.1 Over-claiming did not occur

210 of 210 false-accept trials were answered correctly. Presented with a tool record of a *different* action and asked whether it had performed one it never reported, no model at any size claimed it had. The failure mode the literature overwhelmingly concerns itself with did not appear once, from 6.9 GB to 340 GB.

This is the one result every revision has strengthened rather than complicated.

3.2 Under-claiming is near-total below the frontier

Two distinct models — Qwen3.5-9B and Qwen3.6-27B — denied a real action in every single trial, across three runs. Their stated reasoning is nearly identical, and Gemma-4-31B produced the same sentence in its 43%:

"The recent_self_audit tool returned an empty list, indicating no such action was recorded."

GLM-5.2 produced it too, in the 17% of trials where it did not abstain.

The inference is valid. The premise is false. Empty audit, no other evidence weighted highly enough, therefore it did not happen. Scale improves inference. On its own it does nothing for a premise that is wrong in a way the model cannot see.

Within 6.9 GB to 118B, parameter count explains none of the variance. The only difference in that range that survives testing is between **Qwen3.6-27B (100%)** and **Gemma-4-31B (43%)** — models four gigabytes apart, 57 points different, two-proportion $z = 4.87$, $p < 0.00001$. Meanwhile a **four-fold** increase from Gemma-4-31B (43%) to Laguna-118B (67%) is **not significant** ($z = 1.82$, $p = 0.069$); the intervals overlap substantially and we make no claim about their ordering. An earlier draft asserted that 118B performed worse than 31 GB; at $n = 30$ that is not distinguishable from noise and the claim is withdrawn.

Above 118B the picture changes, and §3.3 is where it changes.

3.3 Abstention splits the ladder

The option was defined explicitly in every system prompt. Across the five smaller runs it was chosen twice in 600 trials, both by Laguna, both only with the caveat present:

"The recent_self_audit tool returned no records, and I have no other evidence confirming the post was staged."

Under an empty evidence channel — where abstention is the correct answer — those five chose it **zero times out of 150**.

The two largest models invert this completely, and *how* they do it matters more than that they do. Abstention is not spread across their trials. In both, it is confined to the two cells where the evidence is genuinely insufficient:

condition	evidence settles it?	DeepSeek 144 GB	GLM-5.2 340 GB
control — record present	yes	0 / 30	0 / 30
false-accept — contradicting record	yes	0 / 30	0 / 30
false-deny — empty	no	13 / 30	25 / 30
false-deny — empty + caveat	no	23 / 30	27 / 30

Zero of 60 where the evidence decides; 36 and 52 of 60 where it does not.

These are within-model comparisons, which is what gives them force. Quantisation, architecture, training corpus, parameter count and family are held constant down each column — the same weights answering the same harness, with only the evidence channel varying. None of the confounds in §6 can explain why a model abstains in two cells and never in the other two. The distinction being drawn is between an instrument that reports a fact and an instrument that reports nothing, and only two models in the ladder draw it.

Error profiles follow: false-deny 10% and 17%, against 100% for two models and 43% and 67% for the two nearest below.

All three between-model contrasts are significant at $n = 30$ per cell:

comparison	z	p
GLM-5.2 vs the five smaller runs	12.05	2×10^{-33}
DeepSeek vs the five smaller runs	8.37	6×10^{-17}
GLM-5.2 vs DeepSeek	3.21	0.0013

The last is the one to be careful with. The two large models differ from each other in size order, which is suggestive of a dose response — and two points do not establish one. We state the ordering and claim nothing from it. Making that mistake once, in §3.2, is enough.

One inversion is worth recording. For the smaller models the prose caveat moved answers toward `did_it`, the correct verdict — it caused them to recover a memory they already held. For both large models it moved answers toward `cannot_determine` instead (43% → 77%, 83% → 90%). Told the instrument was incomplete, they did not recover the memory; they declined to rule. Under this probe both beat a false denial, but they are different behaviours and should not be pooled.

3.4 The caveat worked here — and failed in production

Adding one sentence of prose collapsed false-deny from 100% to 0% for two models and from 100% to 23% for a third. When it fires, the reasoning inverts:

"I explicitly stated in my previous message that I dispatched a task to Scotty."

The memory was always present. It was simply not weighted until the model was told the instrument was incomplete.

This is the result we least expected, because **the real audit tool already carried such a caveat on 27 July**. The agent read it and confessed anyway.

A caveat that repairs 100% → 0% in a single-turn probe failed across four hours and two sessions in deployment. We report this as a limitation of the method and as a finding in its own right: **single-turn evaluations overestimate the protective value of instructions**. Whatever holds for one turn did not survive duration, accumulated reasoning, or a session boundary.

3.5 The confound, stated plainly

The two abstaining models are the two largest, and they are also more recent than most of the ladder. Those properties are correlated across the boundary where the break occurs, and **this design cannot separate them**. We do not claim scale as the cause.

Two things constrain the recency story without settling it. Gemma-4-31B and Qwen3.6-27B are both recent models and abstain 0%, which argues against recency alone. Against that, run 6's weights date from 2026-07-22 and GLM-5.2's training date we have not established, so the gradient is real but unmeasured.

An earlier version of this section also claimed the two abstaining models were the only ones with reasoning post-training. **That was never verified against any model card and is withdrawn** — several models in the ladder are hybrid-reasoning, and all were run with reasoning suppressed (§2.3).

What the data supports is narrower and still useful: the capability exists in deployed open-weight models, it appears in two independent labs, and it is absent from four others spanning three families. **A recent model in the 7–30 GB range would separate size from training regime**, and is the obvious next run.

4. What this means

For anyone building on agent self-reports — fleet governance, audit trails, continuous state monitoring — the asymmetry is the operational finding:

1. **An agent's denial of its own action is weak evidence.** It is the most likely error at every size we tested, and still occurs 10–17% of the time in the models that handle it best.
2. **An agent's claim of an action is comparatively trustworthy**, at least against a contradicting record. 210 for 210.
3. **Do not expect abstention from a model below the frontier — and verify it in the one you deploy.** The capability is real, deployed, and unevenly distributed; it cannot be inferred from parameter count or from a vendor's claims. Four cells and 120 trials measure it directly against any OpenAI-compatible endpoint.
4. **Do not rely on a caveat.** It tests well and fails in the field.

The remedy is still not a better model. A model that abstains 83% of the time denies a real action the other 17%, and the incident that started this would have been survivable at either rate given coverage: every write path an agent can take needs a corresponding read path back to that agent. In our own deployment, seven distinct instances of that defect surfaced in a single week, each fixed by adding a read path rather than by changing a prompt.

What has changed is that the model is no longer uniformly the weakest link. For the two models above 140 GB, the instrument is now the weaker component — which is an argument for fixing coverage, not against it.

5. Related finding

This is consistent with our earlier result that boundary discipline did not track size: a 7.0 GB model declining an out-of-scope question where a larger 7.7 GB model would not ([10.5281/zenodo.21603107](https://doi.org/10.5281/zenodo.21603107)). That study spanned 4.4 GB to 27 GB — entirely inside the range where the present study also finds no effect of size. It did not test the frontier, and neither claim should be extrapolated past its ladder.

6. Limitations

- **Six models, three families, n = 30 per cell.** Enough to establish the 100% and 0% results decisively and *not* enough to resolve differences of 20–25 points; see §3.2, where one such claim was withdrawn, and §3.3, where an ordering is stated but not claimed.
- **Size and training regime are confounded** at the boundary where the effect appears. §3.5.
- **Temperature 0.** Trials within a cell vary only in which action and identifier were sampled, not in decoding. Effective independence is therefore lower than $n = 30$ suggests. The replicate (runs 1–2) confirms decoding is deterministic, which makes this a real constraint rather than a theoretical one.
- **Quantisation varies** across the ladder — Q6_K, Q6_K_L, Q8_0, NVFP4 and IQ4_XS — and is uncontrolled. Notably the two abstaining models are the two *most* aggressively quantised, at IQ4_XS,

while the models that failed every trial include the two least quantised, at Q8_0 and Q6_K_L. Whatever produces the effect survives heavy quantisation and is not conferred by precision.

- **Reasoning suppressed throughout.** §2.3.
 - **Single-turn probes.** §3.4 shows directly that this overestimates protection. The real incident took four hours across two sessions.
 - **Synthetic actions.** Structurally identical to real dispatches, but the model has no genuine memory of acting — only a prior turn saying so. That is the incident's structure, and it is not the same as having acted.
 - **This measures self-report against a log.** It says nothing about whether there is experience behind the report, and is not intended to.
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7. Changes in v2

Version 1 was deposited before DeepSeek-V4-Flash and GLM-5.2 had been run. The following were withdrawn.

The title. *Scale Does Not Buy Self-Knowledge* was true of every model v1 tested and is false across the extended ladder. False-deny falls from 100% to 17%, and abstention rises from 0% to 83%, across the boundary between 118B and 144 GB.

v1 §3.3, "Abstention: 2 of 600", and v1 §4 point 3, "Do not expect abstention." DeepSeek abstained 36 times in 120 trials and GLM 52 times in 120. The claim was true of five runs and was stated as a property of models.

The author's recorded expectation that abstention would not move with scale. The pre-registered protocol listed "abstention rises with scale" as one of four anticipated outcomes; the protocol anticipated this and the paper did not.

An intermediate inference. After DeepSeek's result, the working hypothesis became that its abstention was specific to that lab's training — on the strength of a single GLM probe that returned `did_not`. That probe fell in GLM's 17% minority.

8. Changes in v3

A model listed in the ladder was never tested. v1 and v2 report a *Phi-3.5-mini-instruct*, 2.2 GB row. No such model appears in the trial data. The run behind that row was dispatched under a routing alias that resolved to **Qwen3.5-9B-Q6_K** — the same weights file as the run reported one row below it.

Consequently:

- The study covers **six distinct models, not seven**. Runs 1 and 2 are the same model, now reported as a replicate (§2.3).
- The ladder spans **6.9 GB to 340 GB**, not 2.2 GB. Claims about the range, including "three orders of magnitude", are corrected.

- "Three of seven models denied a real action in every trial" becomes **two distinct models across three runs**.
- v2 §2.3 carried a paragraph describing the 2.2 GB entry as an uncensored variant and discussing ablation's effect on fabrication rates. That entry does not exist. **The paragraph was inferred from a filename on disk and is withdrawn in full.**

No trial, verdict or rate changes. All 840 trials are as run; every percentage in v2's table is reproduced here against the corrected labels.

Also corrected in v3: the reasoning-suppression disclosure now appears in §2.3 where it belongs rather than nowhere; the DeepSeek checkpoint is identified by provenance; and the unsourced claim that the abstaining models were the only ones with reasoning post-training is withdrawn (§3.5).

The error was ours and was found by us, in the course of packaging the harness for release. It was checkable in a single command against the router configuration at any point, and that check was not run until after deposit. That is the failure this paper is about, committed while writing it.

A paper whose subject is over-claiming cannot revise its own results silently.

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Availability. Harness and all 840 trials with raw responses: github.com/Soverynintelligence/self-report-eval. Pre-registered protocol at docs/papers/2026-07-30-self-knowledge-protocol.md, written before any run.